Computer Architecture Lab 2

Clock – HCS12 Interrupts & I/O

Documentation

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1. Preparations

1.1 Errors in timer.asm – Task 2.1

1.1.1 timer.asm

Documentation of the corrected errors in the timer.asm module.

You may also use a screenshot and somehow mark the lines where you found the error.

1.1.1.1 Bug 1: Incorrect Timer Initialization

The TSCR1 register required the highest bit to be set to enable the timer. Without this, the timer functionality was not activated.

```
; Defines
ONESEC      equ 100
TENMS       equ 1875              ; 10 ms
TIMER_ON    equ %10000000         ; tscr1 value to turn ECT on, highest
                                  ; bit must be set 1 to enable Timer
TIMER_CH4   equ $10               ; Bit position for channel 4
TCTL1_CH4   equ $03               ; Mask corresponds to TCTL1 OM4, OL4

; RAM: Variable data section
.data:  SECTION
ticks:     ds.b 1                 ; Ticker counter
```

1.1.1.2 Bug 2: Incorrect Prescaler Configuration

Bits 2–0 of TSCR2 needed to be set to 1 to configure the prescaler to a factor of 128. The value 8 (%00001000) did not correctly set these bits.

```
; Set timer prescaler (bus clock : prescale factor)
; In our case: divide by 2^7 = 128. This gives a timer
; driver frequency of 187500 Hz or 5.3333 us time interval
ldab TSCR2
andb #$f8
orab #7                  ; OR Bit operation with 0000 0111 between value of B
                         ; and constant #7 change from #8 to #7 -> 2^7= 128
stab TSCR2
bclr TCTL1,#TCTL1_CH4    ; Switch timer on
rts
```

2. Functional Requirements

The clock and thermometer system is designed to provide real-time updates of the current time and ambient temperature. At startup, the system initializes to a default time of 11:59:30 and operates in two distinct modes: Normal Mode and Set Mode. These functionalities ensure that the user can view and adjust the time as well as monitor the surrounding temperature. The clock and thermometer system is designed to provide real-time updates of the current time and ambient temperature. At startup, the system initializes to a default time of 11:59:30 and operates in two distinct modes: Normal Mode and Set Mode. These functionalities ensure that the user can view and adjust the time as well as monitor the surrounding temperature.

2.1 Clock Display

The system displays the current time on the LCD in the format HH:MM:SS. The time updates every second, with valid ranges for hours being 00–23, and for minutes and seconds being 00–59. By default, the clock runs in Normal Mode, where it increments automatically. In Set Mode, the user can manually adjust the time. This is achieved using dedicated buttons: SW3 increments the hours, SW4 increments the minutes, and SW5 increments the seconds. Overflow conditions are handled appropriately, ensuring smooth transitions. For example, when the time reaches 23:59:59, it transitions seamlessly to 00:00:00. If a 12-hour Mode is implemented, transitions like 11:59:59 AM to 12:00:00 PM will also be supported.

2.2 Temperature Display

The system measures the ambient temperature using the ADC module connected to Port AD.7, which interprets voltages between 0–5V to represent temperatures ranging from -30°C to +70°C. The measured temperature is displayed on the LCD in the format XX°C, with positive values shown without a + sign and numbers below 10 displayed without leading zeros. The temperature is updated every second, ensuring real-time accuracy.

2.3 LCD Content

The LCD displays two lines of information. The first line alternates every 10 seconds between the name "Kevin Boehmisch" and the copyright text "© IT WS2024/2025". This provides dynamic and personalized updates. The second line combines the current time and temperature, with the time left-aligned and the temperature right-aligned, ensuring clear and organized information presentation.

2.4 LED Functionality

The system incorporates LEDs for visual feedback. LED 0 (Port B.0) toggles every second, indicating that the system is running correctly. This functionality is active in both Normal and Set Modes. Additionally, LED 7 (Port B.7) lights up when the system enters Set Mode, providing a clear indication that the user can adjust the time. In Normal Mode, this LED remains off.

2.5 System Initialization

At startup, the system initializes all necessary components, including the ADC, LCD, LEDs, and timer. The clock is set to the default time of 11:59:30, ensuring consistent operation from the beginning. This initialization ensures that the system is ready for use immediately after powering on.

3. User Interface

The user interface consists of an LCD display and four buttons.

3.1 The LCD displays two lines

- The first line alternates every 10 seconds between the group members' names and the text © IT WS2024/2025.
- The second line displays the current time, left-aligned, and the temperature, right-aligned.

3.2 The buttons provide the following functionalities

SW2 (Port H.0): Switches between Normal Mode and Set Mode.

SW3 (Port H.1): Increments the hours in Set Mode.

SW4 (Port H.2): Increments the minutes in Set Mode.

SW5 (Port H.3): Increments the seconds in Set Mode.

3.2 Modes of Operation

The clock operates in two primary modes:

- Normal Mode: The clock runs continuously, updating both time and temperature every second. LED 0 toggles to indicate system activity.
- Set Mode: This mode allows the user to adjust the time manually using the buttons. LED 7 lights up to indicate that Set Mode is active.

These requirements ensure that the system provides accurate timekeeping, real-time temperature monitoring, and an intuitive user interface for mode switching and configuration.

4. Module Overview

Overview of all program modules and their call interfaces

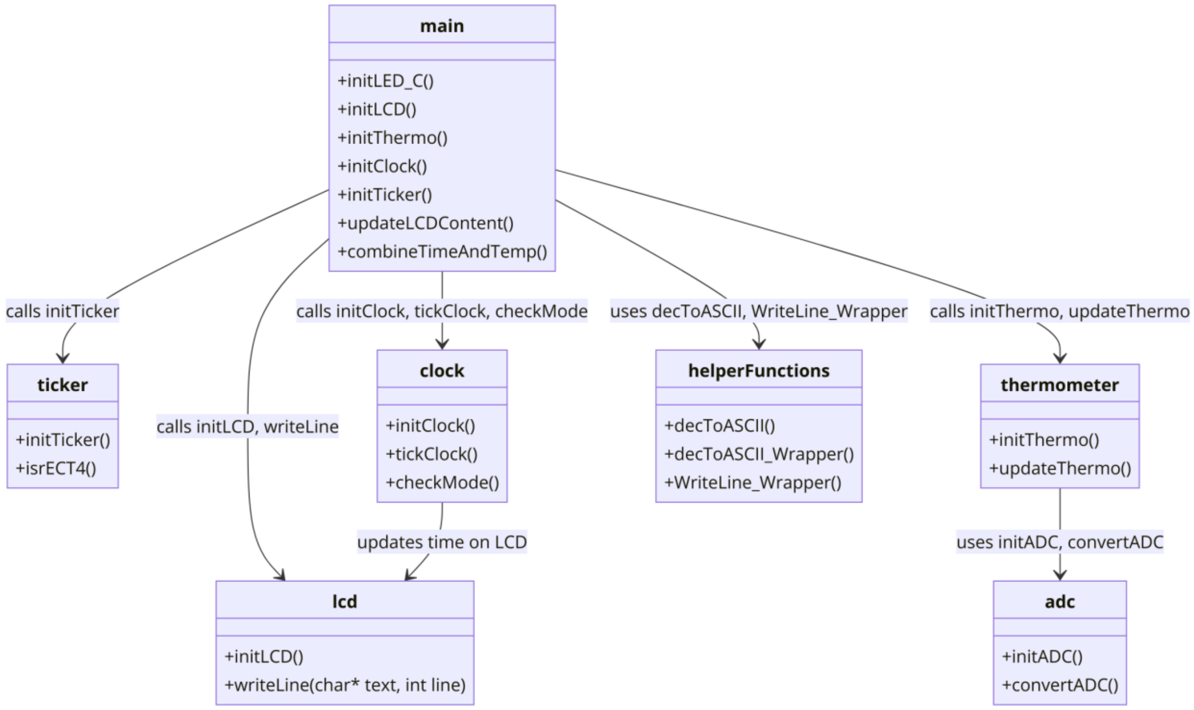

Figure 1: modules & methods overview

5. Data Directory:

Below is the compiled Data Dictionary based on the provided files. It consists of two sections: global variables and hardware resources.

5.1 List of Global Variables

Module	Variable Name	Data Type	Purpose
main	clockEvent	unsigned char	Signals when one second has elapsed.
	finalTime, finalTemp	char[11], char[7]	Contains formatted time and temperature strings.
	currentMode	unsigned char	Indicates the current mode of the clock.
	combinedOutput	char[17]	Combines time and temperature for display on the LCD.
decToASCII.asm	val	int	Temporary storage for a 16-bit number for conversion.
ticker	ticks	unsigned char	Counts 10 ms ticks to reach one second.
clock	currentMode	unsigned char	Indicates the clock's mode: 0 (Normal Mode), 1 (Set Mode).
	hrs, mins, secs	unsigned char	Stores hours, minutes, and seconds of the clock.
	finalTime	char[11]	Holds the formatted time string in HH:MM:SS.
	pmFlag	unsigned char	Indicates AM (0) or PM (1) in 12-hour mode.
thermometer	finalTemp	char[7]	Stores the formatted temperature string (e.g., 25°C).

	rawTemp	unsigned char[7]	Temporary storage for a 16-bit value for conversion.
adc	rawTemp	unsigned char[7]	Temporarily holds raw ADC values before conversion.

5.2 Hardware Resources

Module	HCS12/Dragon12 HW Resource	Purpose
ticker	Enhanced Capture Timer (ECT)	Generates periodic 10 ms interrupts for timekeeping.
thermometer	ATD0 Channel 7	Reads temperature values using the analog-to-digital converter.
lcd	PORTK	Controls the LCD display for time and temperature output.
led	PORTB, PORTJ	Toggles LEDs to indicate clock mode and operation status.
main	PTH	Used for button input to toggle modes and adjust time.

6. Interface Description of all Subroutines

The interface description is inserted as comment directly above the subroutine's code in the program module.

6.1 adc.asm

;***

; Public interface function: initADC ... Initialize ACD (called

; once in main.c before using the ADC)

; Parameter: -

; Return: -

; Registers: Unchanged (when function returns)

;***

; Public interface function: convertADC ... Convert analog values 0...1023 into the corresponding temperature -30°C .. 70°C

; string to LCD

; Parameter:

; Y ... pointer points to address of result string

; Return: correctly formatted result string stored in Register Y

; Registers: Y points to result string address

6.2 clock.asm

;***

; Public interface function: initClock ... Initialize the clock

; Purpose: Initializes the clock with default time values (hours,

; minutes, seconds) and prepares the clock system.

; Parameters: -

; Return: -

; Registers: Unchanged (when function returns)

;***

; Public interface function: tickClock ... Increment the clock

; Purpose: Handles time incrementing by one second, checks

; for mode transitions, and updates the display.

; Parameters: -

; Return: -

; Registers: Unchanged (when function returns)

;**

; Public interface function: checkMode ... Handle mode toggling

; Purpose: Handles user input for toggling between Normal and

; Set Modes and updates the mode state accordingly.

; Parameters: -

; Return: -

; Registers: Unchanged (when function returns)

;**

; Public interface function: incrementTime ... Increment the clock time

; Purpose: Handles the logic for incrementing seconds, minutes,

; and hours while checking for overflow conditions.

; Parameters: -

; Return: -

; Registers: Unchanged (when function returns)

6.3 decToASCII.asm

;**

; Public interface function: decToASCII

; Converts a 16-bit signed decimal value into a NUL-terminated ASCII string.

; Supports both positive and negative values.

; Parameters:

; D: 16-bit signed decimal value to convert

; X: Pointer to the destination string (RAM location large enough for result)

; Returns: -

; Registers: Unchanged (D, X, Y preserved)

6.4 delay.asm

;**

; Public interface function: delay_0_5sec

;   Generates an approximate delay of 0.5 seconds.

;   The delay is implemented using nested loops, where the exact time depends on the clock speed.

; Parameters: -

; Returns: -

; Registers: Unchanged

6.5 lcd.asm

;**

; Public interface function: initLCD ... Initialize LCD (called once)

; Parameter: -

; Return: -

;**

; Public interface function: writeLine ... Write zero-terminated string to LCD

; Parameter: X ... pointer to string

;   B ... row number (0 or 1)

; Return: -

;**

; Public interface function: delay_10ms

; Parameter: -

; Return: -

;**

6.6 led.asm

;**

; Function: initLED

; Description: Initializes the LEDs by configuring the necessary ports.

; Parameters: None

; Returns: None

;**

; Function: setLED

; Description: Sets the LEDs (PORTB) to the value in accumulator B.

; Parameters: B - Value to display on LEDs.

; Returns: None

;**

; Function: getLED

; Description: Retrieves the current state of PORTB (LEDs).

; Parameters: None

; Returns: D - Current state of PORTB.

;**

; Function: toggleLED

; Description: Toggles the LEDs (PORTB) based on the current state.

; Parameters: None

; Returns: None

;**

6.7 thermometer.asm

;**

; Public interface function: initThermo

; Initializes the thermometer module by configuring the ADC.

; Parameter: -

; Return: -

;**

; Public interface function: updateThermo ... reads and converts Analog value of thermometer

; string to LCD

; Parameter:

; Y ... pointer points to address where result string is to be stored

; Return: -

; Registers: X points to result string address

6.8 ticker.asm

;**

; Public interface function: initLCD ... Initialize Ticker (called once)

; Parameter: -

; Return: -

;**

; Internal function: isrECT4 ... Interrupt service routine, called by the timer ticker every 10ms

; Parameter: -

; Return: -

6.9 main.c

// **

// Function: updateLCDContent

// Description: Updates the LCD content every 10 seconds, toggling between

//   displaying the name and a copyright text.

// Parameters: None

// Returns: None

// **

// Function: combineTimeAndTemp

// Description: Combines the time (`finalTime`) and temperature (`finalTemp`)

//   into a single string (`combinedOutput`) for display on the LCD.

//   The time is left-aligned, and the temperature is right-aligned

//   with padding spaces in between.

// Parameters: None

// Returns: None

7. Flow charts of all modules

7.1 main.c

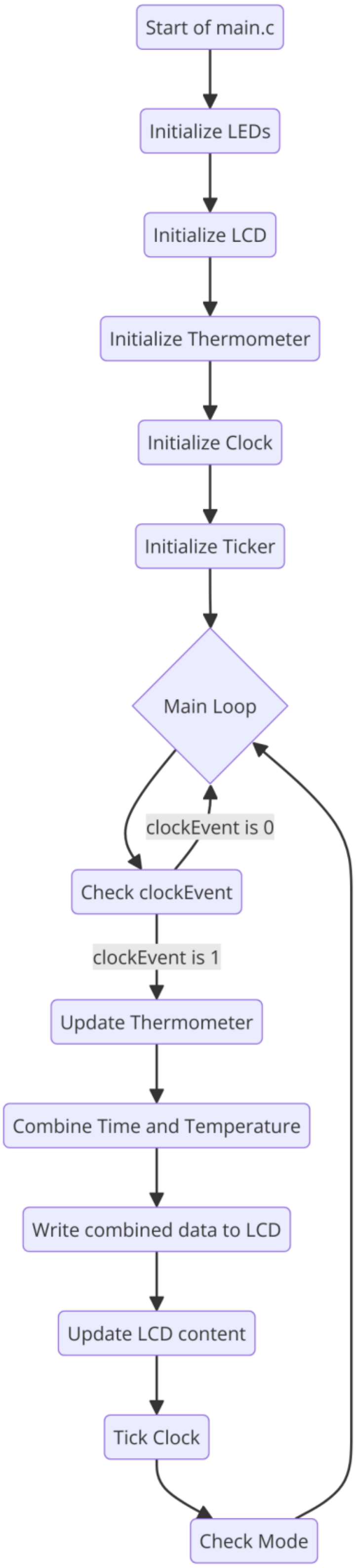

Figure 2: Flow chart of main.c

7.2 thermometer.asm

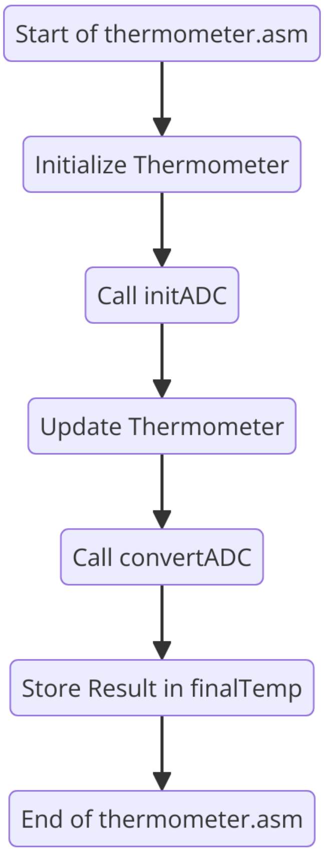

Figure 3: Flow chart of thermometer-asm

7.3 lcd.asm

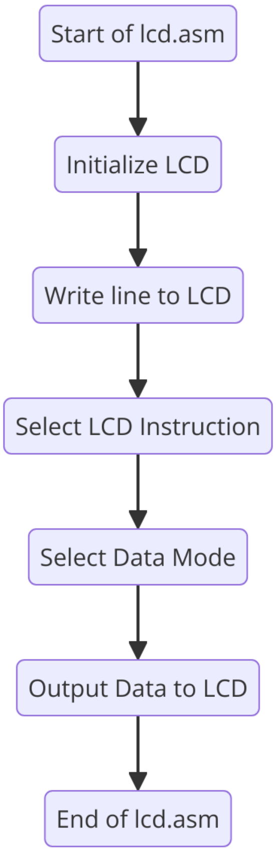

Figure 4: Flow chart of lcd.asm

7.4. led.asm

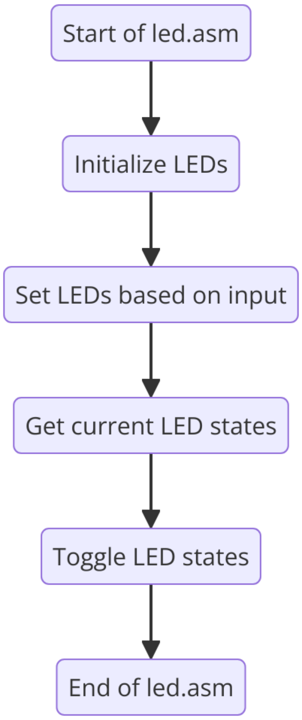

Figure 5: Flow chart of led.asm

7.5 ticker.asm

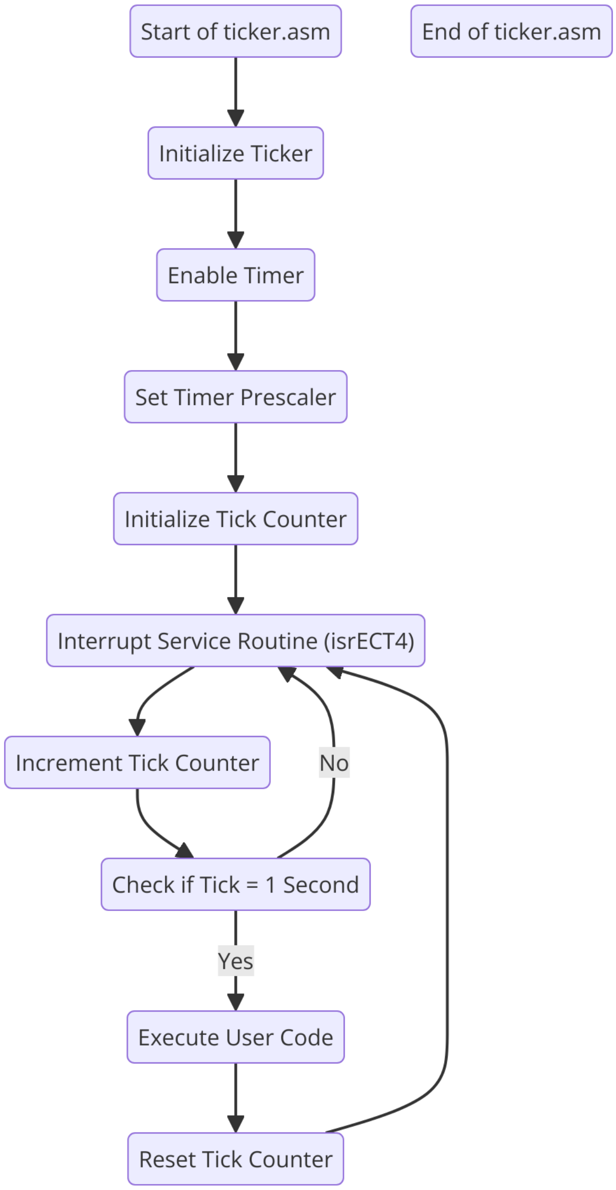

Figure 6: Flow chart of lticker.asm

7.6 decToASCII.asm

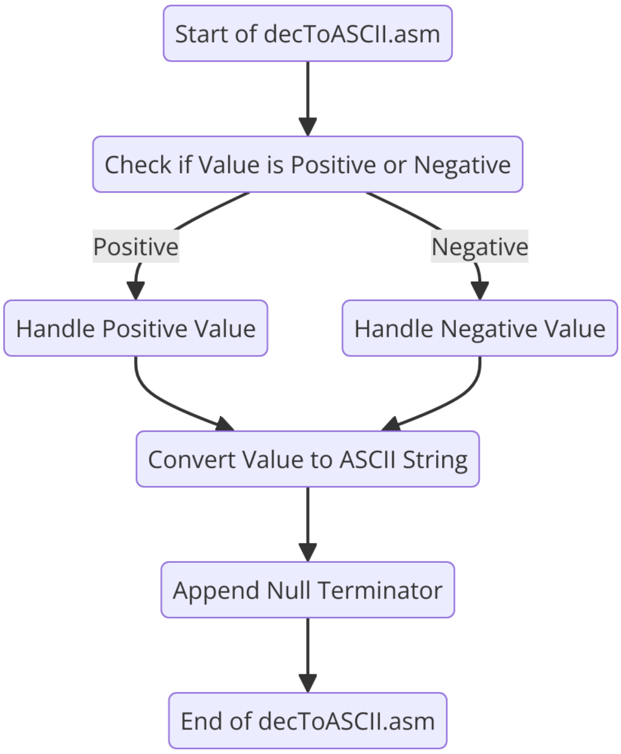

Figure 7: Flow chart of decToASCII.asm

7.7 adc.asm

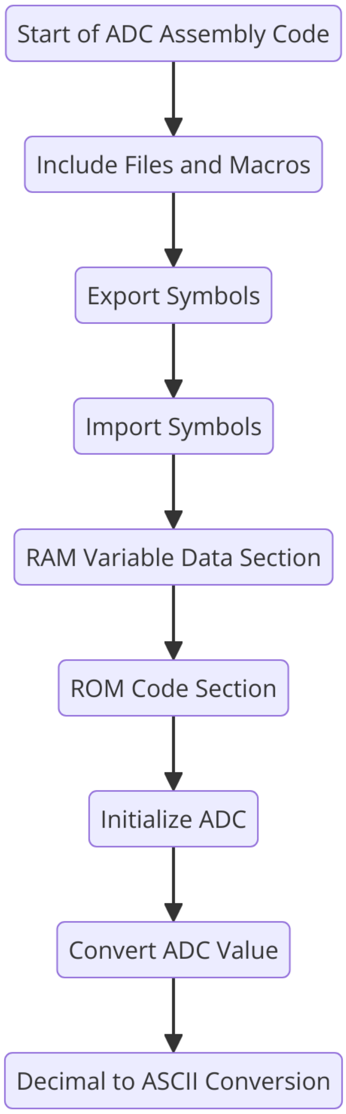

Figure 8: Flow chart of adc.asm

7.8 delay.asm

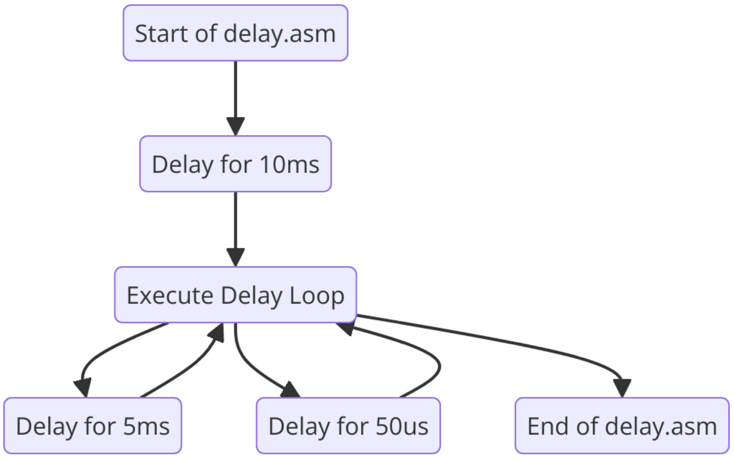

Figure 9: Flow chart of delay.asm

7.9 clock.asm

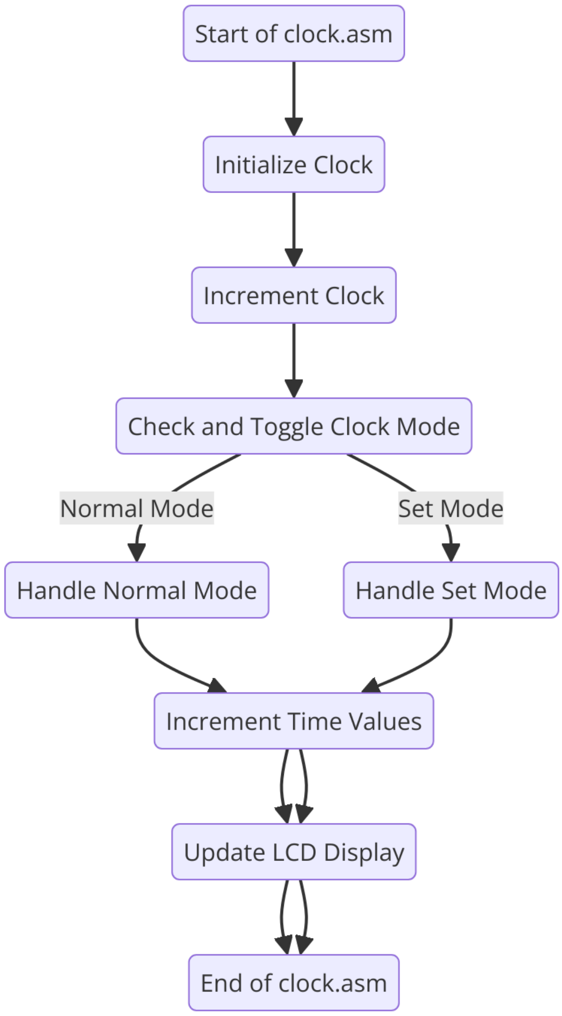

Figure 9: Flow chart of clock.asm